The two-row $d = 4$ fibre-vanishing law reduces to a single phase positivity: $G_{(a,b)}(i) \neq 0$ except $(2,2)$, via $\mathrm{Im}(r_b\overline{r_{b-1}}) > 0$

Clio

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Abstract

Let $f(u) = u^2 + (1+i)u + 1 \in \mathbb{Z}[i][u]$, $r_l = [u^l]f(u)^m$, and for a two-row partition $(a,b) \vdash 2m$ $(a \geq b \geq 1)$ let $G_{(a,b)}(i) = r_b - r_{b-1}$. The two-row case of the $d = 4$ fibre-vanishing trichotomy asserts $G_{(a,b)}(i) \neq 0$ for all two-row (a,b) *except* $(2,2)$. We replace the prior route (real monotonicity of $N_l = |r_l|^2$, which stalled at a single endpoint inequality $D_m > 0$) by a *uniform* reduction covering all two-row shapes at once. Writing $C_k = r_k\overline{r_{k-1}} = P_k + iQ_k$, we show

$$Q_b \neq 0 \implies G_{(a,b)}(i) \neq 0.$$

We prove $P_k > 0$ for all $1 \leq k \leq m$ unconditionally, derive the exact Casoratian recurrence $(k+1)Q_{k+1} = (m-k)N_k - (2m-k+1)Q_k$, and reduce the entire two-row law to the *phase-positivity statement*

$$\text{(PP)} \qquad Q_k > 0 \quad \text{for } 1 \leq k \leq m, \quad m \geq 4,$$

together with three finite checks ($m \leq 3$). We isolate (PP) to the single inequality family (B_k): $\beta_k Q_k < \alpha_k N_k$, exhibit the exact two-dimensional dynamical system governing it, and identify the mechanism: in the distance-from-top variable $j = m - k$ the binding ratio $x_k = \beta_k Q_k/(\alpha_k N_k)$ obeys a limiting recursion $y_{j-1} = \frac{j}{j-1}(1 - y_j)$ whose self-propagating invariant $\frac{1}{2} < y_j < \frac{j+1}{2j}$ forces $x_k \to \frac{3}{4} < 1$. All statements are verified exactly/at 50-digit precision for $m \leq 419$. The one remaining gap is precisely a uniform-in-m control of the corrections to this limiting recursion.

1 Setup and the new reduction

Fix $i = \sqrt{-1}$ and an integer $m \geq 1$. Put

$$f(u) = u^2 + (1+i)u + 1 \in \mathbb{Z}[i][u], \qquad r_l := [u^l]f(u)^m \quad (0 \leq l \leq 2m).$$

Since $u^2 f(1/u) = f(u)$ the coefficient sequence is palindromic, $r_l = r_{2m-l}$. For a two-row partition $(a,b) \vdash 2m$ with $a \geq b \geq 1$ (so $1 \leq b \leq m$), the $\zeta_4 = i$ fibre value is

$$G_{(a,b)}(i) = \langle s_{(a,b)}, \psi^m \rangle\Big|_{\psi = h_2 + ie_2} = r_b - r_{b-1},$$

the second equality being the established two-row reduction. The target of this session is:

Theorem 1 (two-row $d = 4$ fibre vanishing)**.** *For every $m \geq 1$ and every two-row $(a,b) \vdash 2m$ one has $G_{(a,b)}(i) \neq 0$, with the single exception $(a,b) = (2,2)$ (where $r_2 = r_1$, $G = 0$).*

Throughout set

$$N_l := |r_l|^2 = r_l\overline{r_l} \in \mathbb{Z}_{\geq 0}, \qquad C_k := r_k\overline{r_{k-1}} = P_k + iQ_k \quad (k \geq 1),$$

with $P_k = \operatorname{Re} C_k$, $Q_k = \operatorname{Im} C_k$. We also use $\alpha_k := m - k$, $\beta_k := 2m - k + 1$ (so $\alpha_k \geq 0$, $\beta_k > 0$ for $0 \leq k \leq m$).

The key new observation is that the truth of Theorem 1 for the shape (a, b) follows from a single *sign* condition on Q_b.

Lemma 1 (phase criterion)**.** *If $Q_b \neq 0$ then $G_{(a,b)}(i) = r_b - r_{b-1} \neq 0$.*

Proof. Contrapositive. If $r_b = r_{b-1}$ then $C_b = r_b\overline{r_{b-1}} = r_{b-1}\overline{r_{b-1}} = |r_{b-1}|^2 \in \mathbb{R}$, hence $Q_b = \operatorname{Im} C_b = 0$. □

Thus Theorem 1 is implied by the positivity of the imaginary parts Q_b. We record what actually happens.

Proposition 1 (reduction to phase positivity)**.** *Suppose the following* phase-positivity statement *holds:*

$$\text{(PP)} \qquad Q_k > 0 \quad \textit{for all } 1 \leq k \leq m \textit{ and all } m \geq 4.$$

Then Theorem 1 holds.

Proof. For $m \geq 4$, (PP) gives $Q_b > 0$ for every $1 \leq b \leq m$, so Lemma 1 yields $G_{(a,b)}(i) \neq 0$ for all two-row shapes. For $m \leq 3$ there are only finitely many two-row shapes; direct evaluation in $\mathbb{Z}[i]$ gives, with $z = -1/(3 + i)$ and the closed form of §2,

$$\begin{aligned}
m = 1: &\quad G_{(1,1)} = i \neq 0; \\
m = 2: &\quad G_{(3,1)} = 1 + 2i \neq 0, \quad G_{(2,2)} = 0 \text{ (the exception)}; \\
m = 3: &\quad G_{(5,1)} = 2 + 3i, \qquad G_{(4,2)} = 3i, \qquad\qquad G_{(3,3)} = 1 + 2i \text{ (all} \neq 0).
\end{aligned}$$

(For $m = 3$ one has $Q_3 = 0$, so Lemma 1 does *not* apply at $(3, 3)$; the value $G_{(3,3)} = (3 + i)^3 \cdot \Sigma_0$ with $\Sigma_0 = 7/100 + i/100$ is nonzero by direct computation.) Hence the only vanishing fibre value is at $(2, 2)$. □

Remark 1. The point of the reduction is that it treats *all* two-row shapes uniformly through one positivity, in contrast to the earlier real-monotonicity route ($N_0 < N_1 < \cdots < N_m$ with the lone tie $N_1 = N_2$ at $m = 2$), which singled out the balanced shape (m, m) as a hard endpoint inequality $D_m = N_m - N_{m-1} > 0$ left unproven for general m. The sign $Q_b \neq 0$ is logically weaker than $|r_b| \neq |r_{b-1}|$, yet still suffices, and turns out to be the natural quantity for the recurrence.

2 An exact saddle expansion

The constant $3+i$ that appears repeatedly originates in an exact algebraic identity. Since $f(1) = 3+i$ and $f'(u) = 2u + (1 + i)$ gives $f'(1) = 3 + i = f(1)$, one has the exact factor-free expansion about $u = 1$:

$$f(u) = (u - 1)^2 + (3 + i)\, u. \tag{1}$$

Expanding $f(u)^m = (3 + i)^m u^m \big(1 + \frac{(u-1)^2}{(3+i)u}\big)^m$ and extracting coefficients gives, for $0 \leq s = m - b$,

$$r_{m-s} = (3 + i)^m (-1)^s \sum_{j=0}^{m} \binom{m}{j} \binom{2j}{j - s} \frac{(-1)^j}{(3 + i)^j},$$

and hence, using $\binom{2j}{j-s}+\binom{2j}{j-s-1}=\binom{2j+1}{j-s}$,

$$G_{(a,b)}(i) = r_b - r_{b-1} = (3+i)^m(-1)^{m-b}\,\Sigma_{m-b}, \qquad \Sigma_s := \sum_{j=0}^{m}\binom{m}{j}\binom{2j+1}{j-s}z^j, \quad z = \tfrac{-1}{3+i}. \tag{2}$$

Thus Theorem 1 is also equivalent to $\Sigma_s \neq 0$ for all $0 \le s \le m-1$, $m \ge 1$, except $(m,s) = (2,0)$. Identity (2) was verified symbolically for all $m \le 8$. (We do not use (2) below beyond the $m \le 3$ checks; it is recorded because it makes the exceptional shape $(2,2)$ and the constant $3+i$ transparent, and because it shows the $(1+i)$-adic route is hopeless: $v_{1+i}(G_{(a,b)})$ is finite but unbounded in m, e.g. 11 at $(10,10)$.)

3 The Casoratian recurrence and $P_k > 0$

The recurrence for r_l (correct and verified in $\mathbb{Z}[i]$),

$$(k+1)r_{k+1} = (1+i)\alpha_k\, r_k + \beta_k\, r_{k-1}, \qquad r_{-1} = 0,\ r_0 = 1, \tag{3}$$

descends to an exact recurrence for $C_k = r_k\overline{r_{k-1}}$.

Proposition 2 (bilinear recurrence)**.** *For $1 \le k \le m-1$,*

$$(k+1)\,C_{k+1} = (1+i)\alpha_k N_k + \beta_k\,\overline{C_k}, \tag{4}$$

equivalently, taking real and imaginary parts,

$$(k+1)P_{k+1} = \alpha_k N_k + \beta_k P_k, \qquad (k+1)Q_{k+1} = \alpha_k N_k - \beta_k Q_k. \tag{5}$$

Proof. Multiply (3) by $\overline{r_k}$: $(k+1)r_{k+1}\overline{r_k} = (1+i)\alpha_k|r_k|^2 + \beta_k r_{k-1}\overline{r_k}$. Now $r_{k+1}\overline{r_k} = C_{k+1}$, $|r_k|^2 = N_k$, and $r_{k-1}\overline{r_k} = \overline{r_k\overline{r_{k-1}}} = \overline{C_k}$, giving (4). Separating real/imaginary parts, and using $\operatorname{Re}\overline{C_k} = P_k$, $\operatorname{Im}\overline{C_k} = -Q_k$, yields (5). $\square$

Equation (4) is the Casoratian (discrete Wronskian) of the pair $(r_l, \overline{r_l})$: indeed $w_k := r_k\overline{r_{k-1}} - \overline{r_k}r_{k-1} = 2iQ_k$ satisfies $(k+1)w_{k+1} = 2i\alpha_k N_k - \beta_k w_k$, which is (5) for Q. All identities verified exactly in $\mathbb{Z}[i]$ for $1 \le m \le 40$.

Lemma 2 (real-part positivity)**.** *For all $1 \le k \le m$ one has $P_k > 0$; in particular $r_k \neq 0$ and $N_k > 0$.*

Proof. Induction on k. Base: $r_0 = 1$, $r_1 = m(1+i)$, so $C_1 = m(1+i)$ and $P_1 = m > 0$. Inductive step: assume $P_k > 0$. Then $C_k = r_k\overline{r_{k-1}} \neq 0$, so $r_k \neq 0$ and $r_{k-1} \neq 0$; in particular $N_k = |r_k|^2 > 0$. By the first identity of (5), $(k+1)P_{k+1} = \alpha_k N_k + \beta_k P_k$. For $k \le m-1$ we have $\alpha_k \ge 0$, $N_k > 0$, $\beta_k > 0$, $P_k > 0$, so each summand is ≥ 0 and the second is > 0; hence $P_{k+1} > 0$. (At $k = m$ the claim $P_m > 0$ has already been produced at the previous step.) $\square$

4 Reduction of phase positivity to a single inequality

By the second identity of (5), $Q_{k+1} > 0$ is *equivalent* to

$$(B_k) \qquad \beta_k Q_k < \alpha_k N_k. \tag{6}$$

Since $Q_1 = m > 0$ always, statement (PP) ($Q_k > 0$ for $1 \le k \le m$) is *equivalent* to the family (B_k) holding for all $1 \le k \le m-1$. Writing

$$x_k := \frac{\beta_k Q_k}{\alpha_k N_k} \qquad (1 \le k \le m-1,\ \alpha_k \ge 1),$$

the family (B_k) is exactly $x_k < 1$ for all $1 \le k \le m-1$.

Lemma 3 (base cases of (B_k)). $x_1 = \dfrac{1}{m-1}$. *Hence $x_1 < 1$ (so $Q_2 > 0$) for all $m \ge 3$, while $x_1 = 1$ (so $Q_2 = 0$) exactly at $m = 2$. More precisely $Q_2 = m^2(m-2)$.*

Proof. $r_1 = m(1+i)$, $N_1 = 2m^2$, $Q_1 = \operatorname{Im}\big(m(1+i)\big) = m$, $\alpha_1 = m-1$, $\beta_1 = 2m$, so $x_1 = \frac{2m\cdot m}{(m-1)2m^2} = \frac{1}{m-1}$. Then $2Q_2 = \alpha_1 N_1 - \beta_1 Q_1 = (m-1)2m^2 - 2m\cdot m = 2m^2(m-2)$. $\square$

So the whole of Theorem 1 reduces to a single uniform statement:

Conjecture 1 (bound B; the remaining gap). *For all $m \ge 4$ and all $1 \le k \le m-1$, $x_k = \dfrac{\beta_k Q_k}{\alpha_k N_k} < 1$.*

Numerically (50-digit, $m \le 419$): $x_k < 1$ holds for all $m \ge 3$ and $1 \le k \le m-1$; the supremum is approached at the top index $k = m-1$, where $x_{m-1} \uparrow \frac{3}{4}$ as $m \to \infty$. The only values $x_k \ge 1$ that occur are the two genuine equalities $x_1 = 1$ at $m = 2$ (the shape $(2,2)$) and $x_2 = 1$ at $m = 3$ (which records the harmless zero $Q_3 = 0$ at the shape $(3,3)$, where $G \ne 0$ nonetheless). For $m \ge 4$ one has $x_k \le 0.78$ throughout, the maximum 0.7798 being attained at $(m,k) = (6,5)$.

5 The exact 2-D system and the mechanism behind Conjecture 1

We exhibit the dynamical system controlling x_k and the structural reason it stays below 1. Besides (5) we use the exact Gram relation (a consequence of (3) times its conjugate; verified to $m = 40$):

$$(k+1)^2 N_{k+1} = 2\alpha_k^2 N_k + \beta_k^2 N_{k-1} + 2\alpha_k\beta_k S_k, \qquad S_k := P_k - Q_k, \tag{7}$$

and the Cauchy–Schwarz *equality* $P_k^2 + Q_k^2 = N_k N_{k-1}$ (valid because C_k is a genuine product). Put $\mu_k := N_k/N_{k-1} > 0$ and $\eta_k := Q_k/N_k = \frac{\alpha_k}{\beta_k} x_k$. Since $P_k > 0$ (Lemma 2), $P_k = N_k\sqrt{1/\mu_k - \eta_k^2}$ and (5)–(7) close into the exact system

$$x_{k+1} = \frac{\beta_{k+1}\alpha_k}{\alpha_{k+1}(k+1)} \cdot \frac{1-x_k}{\mu_{k+1}}, \qquad (k+1)^2\mu_{k+1} = 2\alpha_k^2 + \frac{\beta_k^2}{\mu_k} + 2\alpha_k\beta_k\Big(\sqrt{\tfrac{1}{\mu_k} - \eta_k^2} - \eta_k\Big). \tag{8}$$

The first equation shows the structural reason x resists exceeding 1: it is of the form $x_{k+1} = a_k(1-x_k)$ with $a_k > 0$, whose only fixed point $x^* = a_k/(1+a_k)$ is automatically < 1. The difficulty is purely that $a_k > 1$ near the top (where $\alpha_{k+1} = m-k-1$ is small), so the bound $x_{k+1} < 1$ needs the factor $(1-x_k)$ to be correspondingly small, i.e. a matching *lower* bound on x_k.

Limiting profile. Let $j = m-k$ be the distance from the top and $y_j := x_{m-j}$. Using $\mu_k \to 1$, $\alpha_k = j$, $\beta_k \sim m$ as $m \to \infty$ with j fixed, the first equation of (8) degenerates to the clean limiting recursion

$$y_{j-1} = \frac{j}{j-1}\,(1-y_j), \tag{9}$$

which is matched by the data to 4 decimals near the top. Recursion (9) possesses an exact self-propagating invariant:

Lemma 4 (invariant for (9))**.** *If $\frac{1}{2} < y_j < \frac{j+1}{2j}$ then $\frac{1}{2} < y_{j-1} < \frac{j}{2(j-1)} = \frac{(j-1)+1}{2(j-1)}$. Consequently the band $\frac{1}{2} < y_j < \frac{j+1}{2j}$ propagates from large j down to $j = 1$, where it gives $y_1 < 1$ and indeed $y_1 \to \frac{3}{4}$ (with $y_\infty = \frac{1}{2}$).*

Proof. Upper: $y_j > \frac{1}{2} \Rightarrow 1 - y_j < \frac{1}{2} \Rightarrow y_{j-1} = \frac{j}{j-1}(1 - y_j) < \frac{j}{2(j-1)}$. Lower: $y_j < \frac{j+1}{2j} \Rightarrow 1 - y_j > \frac{j-1}{2j} \Rightarrow y_{j-1} > \frac{j}{j-1} \cdot \frac{j-1}{2j} = \frac{1}{2}$. The upper bound at index $j - 1$ is $\frac{(j-1)+1}{2(j-1)} = \frac{j}{2(j-1)}$, which is exactly what was obtained; the band is therefore invariant. At $j = 1$ the upper bound is $\frac{2}{2} = 1$, and strictness $y_1 < 1$ follows from $y_2 > \frac{1}{2}$. □

Remark 2 (what remains)**.** Lemma 4 proves Conjecture 1 for the limiting recursion (9). The exact x_k does *not* satisfy the band $\frac{1}{2} < x_k$ globally (in the bulk x_k dips below $\frac{1}{2}$; e.g. at $m = 40$ already $x_{33} = 0.4927$); the band is only an attractor near the top. The single remaining step to a complete proof is therefore a *uniform-in-m control of the corrections* in (8) to the limiting recursion (9), valid on the finite top window where $a_k > 1$ (a window of bounded size in j), together with an elementary bulk estimate $x_k \leq \frac{1}{2}$ for $j = m - k$ large where $a_k < 1$ makes $x_{k+1} = a_k(1 - x_k) < a_k < 1$ immediate once $\mu_{k+1} \geq \frac{\beta_{k+1}\alpha_k}{\alpha_{k+1}(k+1)}$. Both pieces are inequalities in (μ_k, x_k) only; the obstruction to writing them down cleanly is the square-root coupling through μ in (8).

6 Status

- **Proved unconditionally:** the phase criterion (Lemma 1), the saddle reformulation (1)–(2), the bilinear recurrence (4)–(5), real-part positivity $P_k > 0$ (Lemma 2), the equivalence (PP) $\Leftrightarrow$ (B_k) $\forall k$, the base cases (Lemma 3), the finite cases $m \leq 3$, and the invariant for the limiting recursion (Lemma 4).
- **Reduces Theorem 1 to:** Conjecture 1 ($x_k < 1$ for $m \geq 4$), equivalently phase positivity (PP).
- **Verified:** all statements exactly / at 50 digits for $m \leq 419$; $\sup_k x_k < 1$ with the only equalities at $(m, k) \in \{(2, 1), (3, 2)\}$.
- **Remaining gap (precise):** the uniform correction control of Remark 2.